Research defense and next questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Research defense and next questions

Separate descriptive results, computational consistency and untested generalization while proposing a controlled follow-up.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **research defense** and **next question**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Presentation and challenge session. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of presentation and challenge session

Reference example, not the only acceptable wording: Teacher explanation: A presentation should make the **evidence** easy to challenge. Show the baseline, scope, negative cases and uncertainties instead of only a best example.

Use a short sequence:question, design, primary result, failure, limitation, **next question**. Answer unfamiliar challenges by locating **evidence** or stating it is missing.

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of independent reproduction check

Reference example, not the only acceptable wording: Teacher explanation: Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help.

Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of writing the methods section

Reference example, not the only acceptable wording: Teacher explanation: A methods section should let a reader reconstruct the design and computation. Detail should explain choices that affect **evidence**.

Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.

Pictured end goal

Explain **research defense** and **next question**s using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Presentation and challenge session

Independent reproduction check

Writing the methods section

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A **research defense** distinguishes what was learned, what failed and which **next question** is justified. A complete investigation can end with a limited or **negative finding**.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **research defense** and **next question**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

A **research defense** distinguishes what was learned, what failed and which **next question** is justified. A complete investigation can end with a limited or **negative finding**.

Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new **evidence**.

Defense item	Complete supplied exercise contract
Main results	Candidate 12/16; baseline 11/16
Main replay	Matching replay stipulated by the scenario; not newly performed here
Main missing scope	No fresh population sample or user study
Separate picture	Frozen model: clear images 8/10, shaded images 4/10
Picture grouping	Source objects grouped within conditions; number and cross-condition pairing not supplied
Plausible alternatives	Lighting, camera or object differences may matter; cause not established
Proposed follow-up	Same held-out objects under both lighting conditions, one camera, fixed model/scoring

Defense item	Complete supplied exercise contract
Changed follow-up	Camera changes as well as lighting, reintroducing a competing explanation

Main evidence ledger and checked key

Main defense component	Supported statement
Candidate observed rate	12/16 = 75%
Baseline observed rate	11/16 = 68.75%
Descriptive difference	One of sixteen = 6.25 percentage points
Replay matches	Supports recorded computational consistency under the supplied scenario
Fresh population / user benefit	Not studied
Significance / universal usefulness	Not supplied or established
Next work	Resolve a named evidence gap with an explicit new design, not a stronger headline

Main candidate 12/16 = 75%, baseline 11/16 = 68.75%, gain 6.25 percentage points. Matching replay supports computational consistency only; no fresh population/user study. Picture clear 80%, shaded 40%, gap forty points; object-count/pairing details are unknown.

Same held-out objects under both lights with one camera and fixed scoring is a proposed control, not a completed causal test; changing cameras reintroduces an alternative. Independent candidate 9/12 = 75%, baseline 10/12 ≈ 83.333%, candidate difference −1/12 ≈ −8.333 points despite matching reproduction. Preserve the negative result.

A useful next question targets a documented gap, with fresh reserved units and matched criteria; no invented mechanism or future success is required.

Research defense and next questions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study.	Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.	75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: 75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested. Qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study.

75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Defense audit	Checked interpretation
Candidate one more of sixteen	Report 6.25-point descriptive gain with denominator
Replay agrees	Computational consistency, not new-data generalization
No user study	No measured user-benefit claim
Shaded slice lower	A hypothesis about lighting, not established cause
Grouped source objects	Count/independence/pairing still need actual design records
Same-object one-camera proposal	Helps distinguish explanations; not an automatic causal guarantee
Camera and lighting both vary	Confounding remains
Candidate loses in independent case	A reproducible negative result is still a valid research outcome

Separate pictured facts and full reasoning

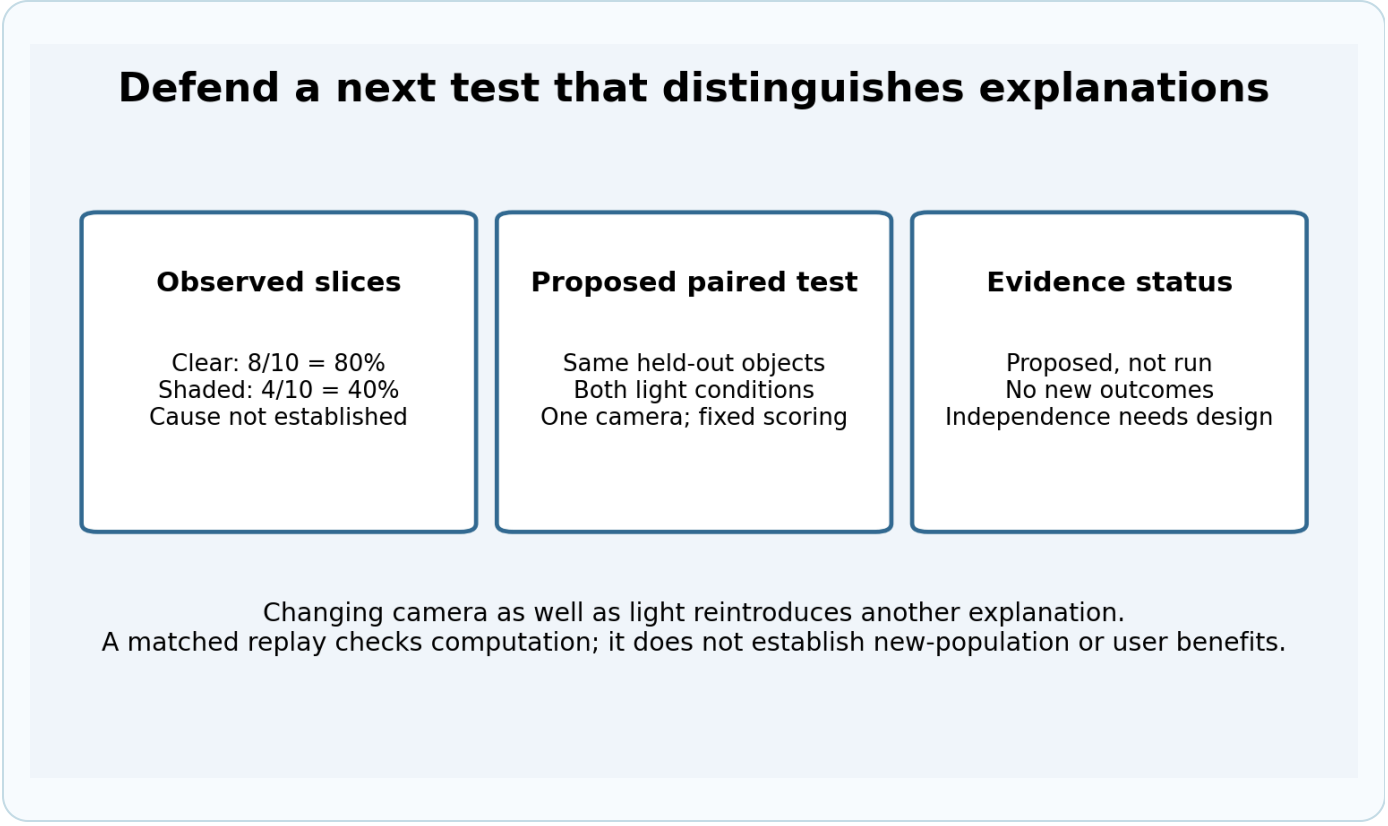
Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.

A frozen model is correct on 8/10 clear images and 4/10 shaded images. Source objects are grouped within each condition.

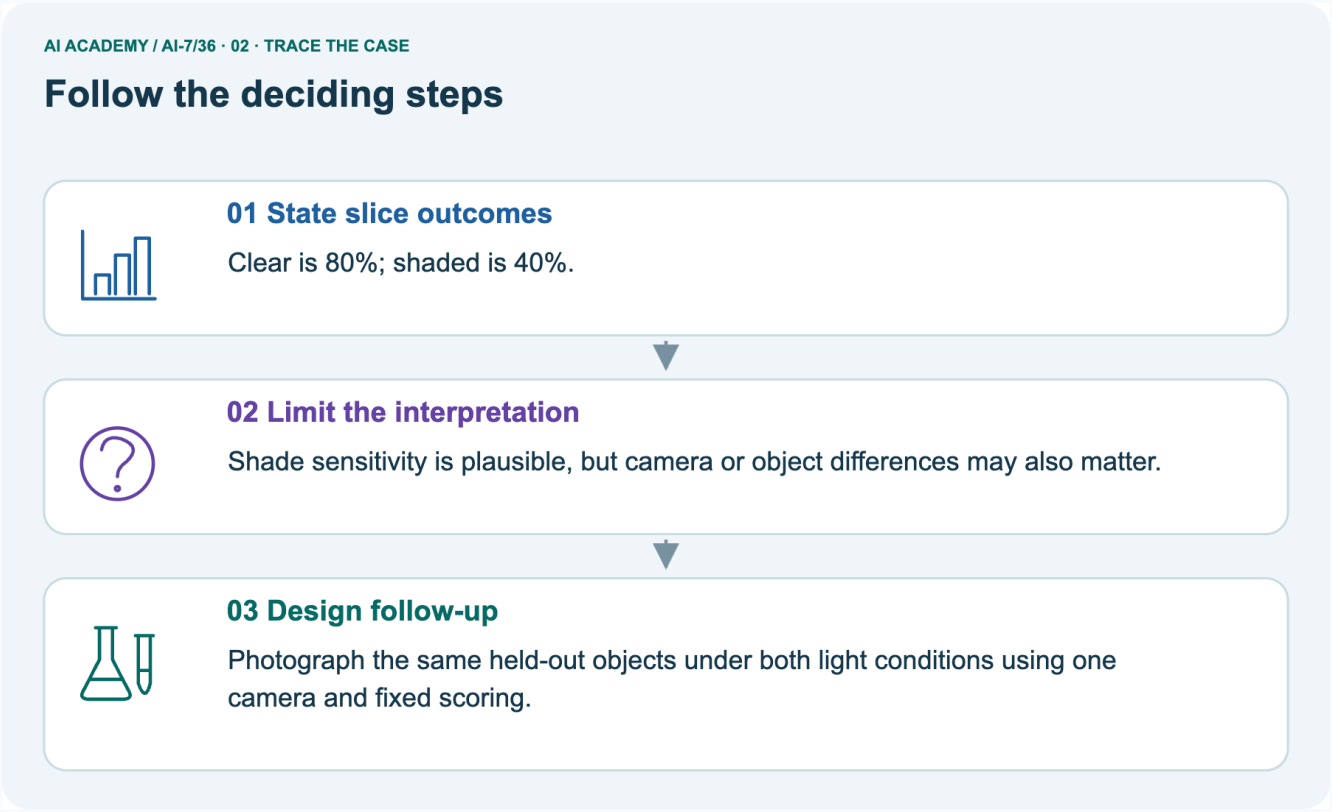
The separate picture gives clear-image correctness eight of ten, or 80%, and shaded-image correctness four of ten, or 40%, a forty-percentage-point descriptive gap. Grouping is mentioned, but the number of source objects and whether the same objects appear in both conditions are not supplied. Lighting sensitivity is plausible; camera and object differences are also possible explanations.

A proposed follow-up photographs the same held-out objects in both light conditions using one camera and fixed scoring. It reduces those particular differences, but still needs adequate independent objects, order/collection controls and justified analysis. No follow-up outcomes or causal finding are supplied. Changing cameras as well as lighting would again entangle explanations.

Separate defense evidence	Supported value or next-step status
Clear slice	8/10 = 80%
Shaded slice	4/10 = 40%
Observed gap	40 percentage points
Object identity/pairing details	Not supplied; no assumption of ten independent objects
Possible explanations	Light, camera or object differences; untested
Proposed matched design	Same held-out objects; both light conditions; one camera; frozen procedure
If camera also changes	Alternative explanation returns
Follow-up status	Not run; no new outcome or causal conclusion



Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State slice outcomes	Clear is 80%; shaded is 40%.
2. Limit the interpretation	Shade sensitivity is plausible, but camera or object differences may also matter.
3. Design follow-up	Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

Change: Change cameras as well as lighting in the follow-up.

Prediction: The design again entangles two conditions.

Reason: A camera difference provides another explanation for any performance gap.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Research defense	A justified account of methods, findings and limits
Next question	A new bounded inquiry motivated by a gap
Negative finding	A result failing the declared expectation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

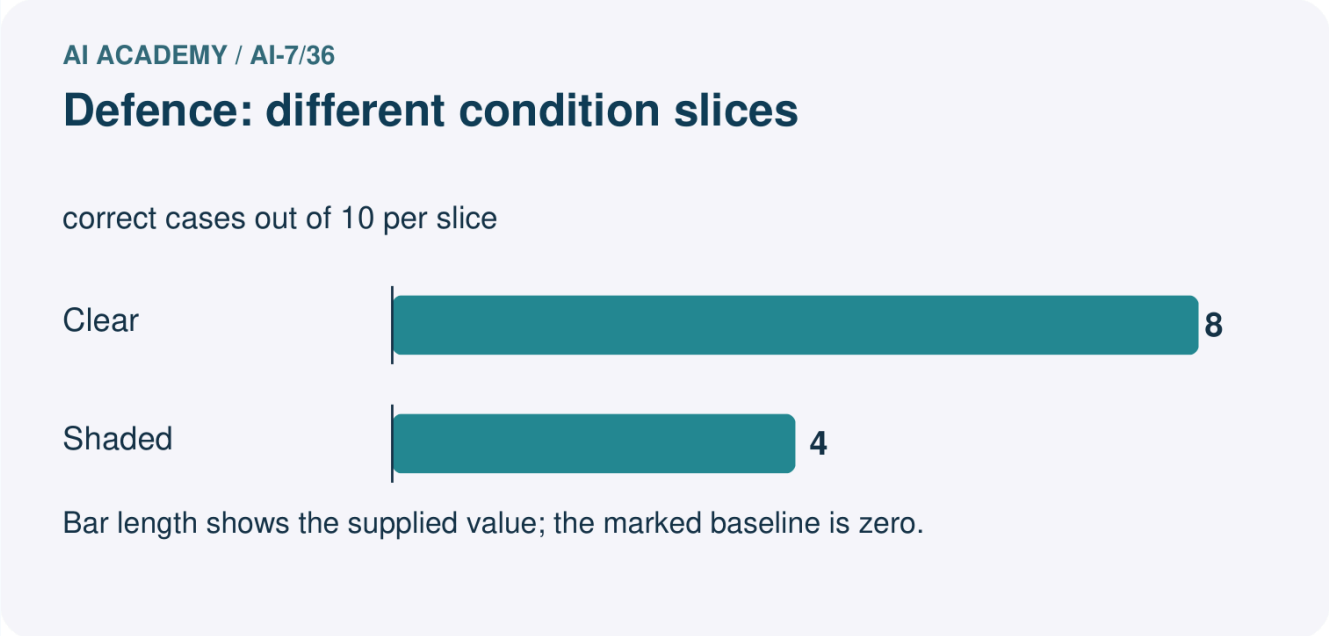
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Research defense and next questions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/36

Research defense and next questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State slice outcomes

Clear is 80%; shaded is 40%.



2. Limit the interpretation

Shade sensitivity is plausible, but camera or object differences may also matter.



3. Design follow-up

Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

CHANGE ONE THING

Change cameras as well as lighting in the follow-up.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: 75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Research defense** and **next question**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

Clear is **80%**; shaded is 40%. Shade sensitivity is plausible, but camera or object differences may also matter. Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Changed case: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Changed-case reasoning: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main candidate 12/16 = 75%, baseline 11/16 = 68.75%, gain 6.25 percentage points. Matching replay supports computational consistency only; no fresh population/user study. Picture clear 80%, shaded 40%, gap forty points; object-count/pairing details are unknown.

Same held-out objects under both lights with one camera and fixed scoring is a proposed control, not a completed causal test; changing cameras reintroduces an alternative. Independent candidate 9/12 = 75%, baseline 10/12 ≈ 83.333%, candidate difference −1/12 ≈ −8.333 points despite matching reproduction. Preserve the negative result.

A useful next question targets a documented gap, with fresh reserved units and matched criteria; no invented mechanism or future success is required.

Diagnose the current misconceptions

The worked result establishes every future case: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**.

Teacher explanation: 75% versus83.333%, candidate lower8.333 points. Investigate recorded failure conditions with a revised prespecified method and fresh reserved cases; do not erase the negative result.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A **research defense** distinguishes what was learned, what failed and which **next question** is justified. A complete investigation can end with a limited or **negative finding**.

Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new **evidence**.

W2 Trace the worked case

Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: 75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

W3 Audit the limit

A report omits this qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

The new case should test this particular **mechanism** rather than merely restate the original result.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: 75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested. Qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**.

Teacher explanation: 75% versus83.333%, candidate lower8.333 points. Investigate recorded failure conditions with a revised prespecified method and fresh reserved cases; do not erase the negative result.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: 75% versus83.333%, candidate lower8.333 points. Investigate recorded failure conditions with a revised prespecified method and fresh reserved cases; do not erase the negative result.

Require intermediate reasoning and the stated scope. Fresh retry: Both procedures8/10 and successful replay. Expected reasoning: Both**80%** in this sample; strict improvement not observed.
</details>

Equivalence is not established without its own design. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Investigate a bounded question with justified baselines, controlled experiments, uncertainty and reproducible **evidence**, then defend the findings and their limitations..

Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **research defense** and **next question**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: 75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Check every learner before discussion: Use the worked case for **Research defense** and **next question**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change cameras as well as lighting in the follow-up. Require a **prediction** and a reason before comparing. Expected result: The design again entangles two conditions.

Reason: A camera difference provides another explanation for any performance gap.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

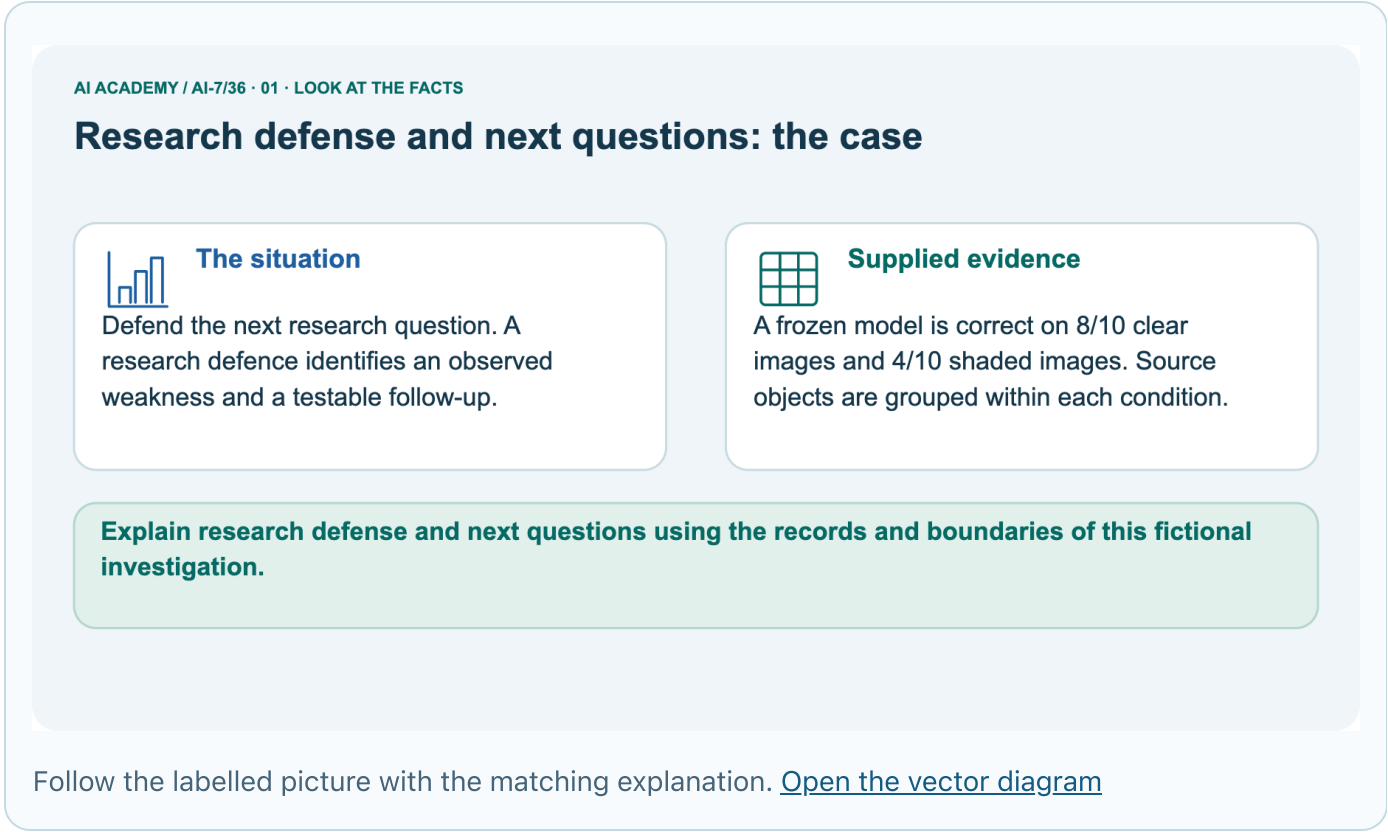
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Research defense and next questions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

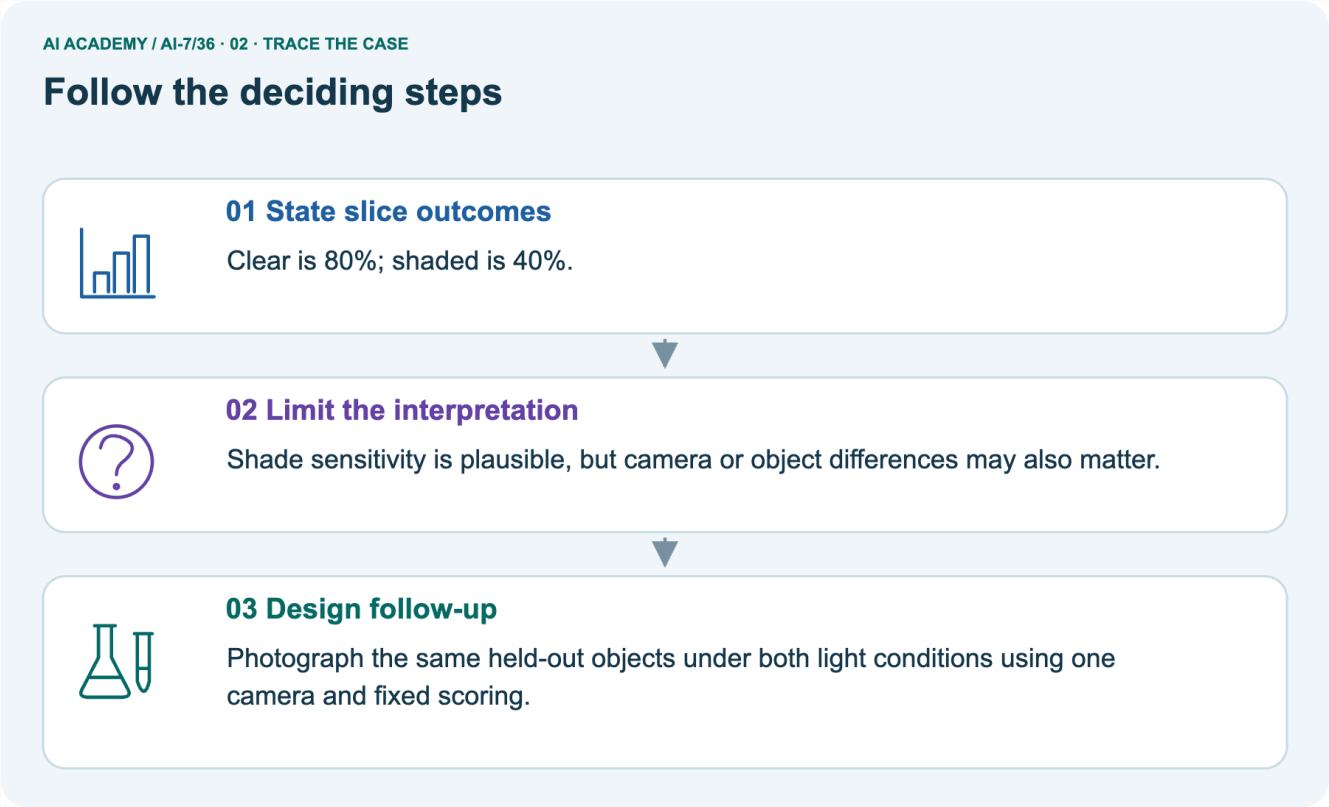
Defend the next research question. A research defence identifies an observed weakness and a testable follow-up. Supplied facts: A frozen model is correct on 8/10 clear images and 4/10 shaded images.

Source objects are grouped within each condition.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



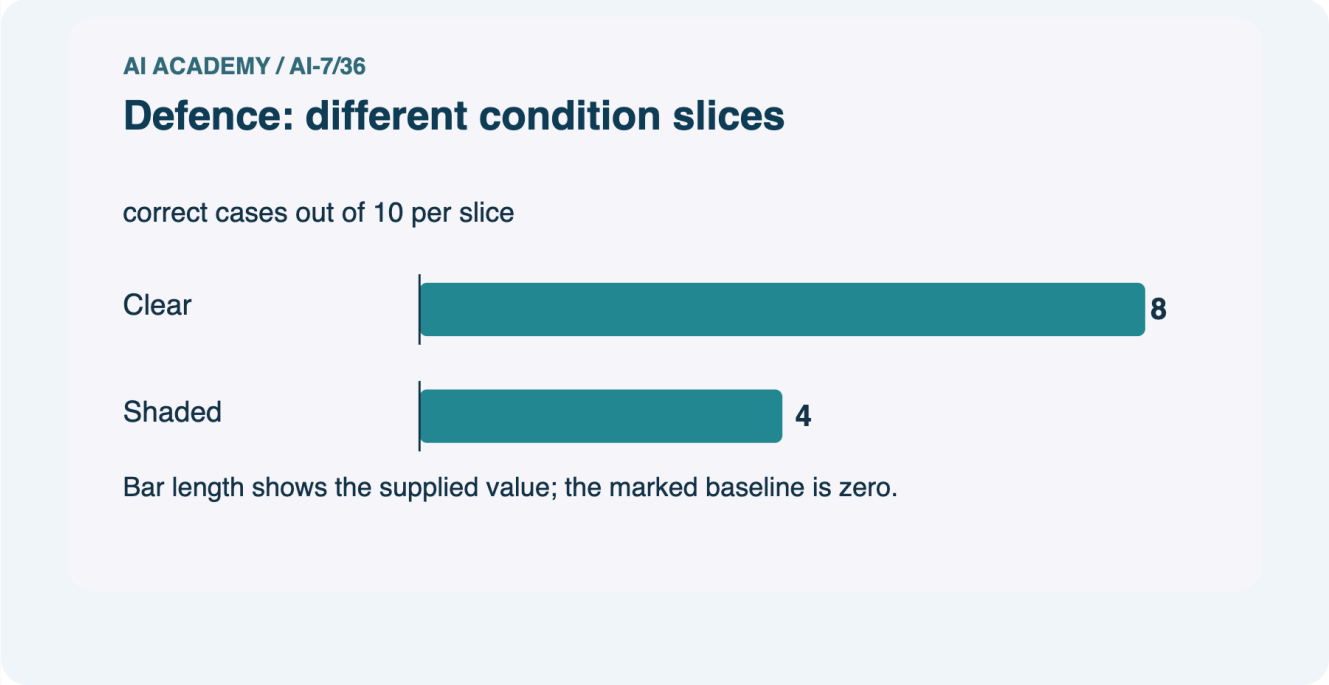
Follow the labelled picture with the matching explanation. Open the vector diagram

State slice outcomes: Clear is **80%**; shaded is 40%. → Limit the interpretation: Shade sensitivity is plausible, but camera or object differences may also matter. → Design follow-up: Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Defence: different condition slices



Follow the labelled picture with the matching explanation. Open the vector diagram

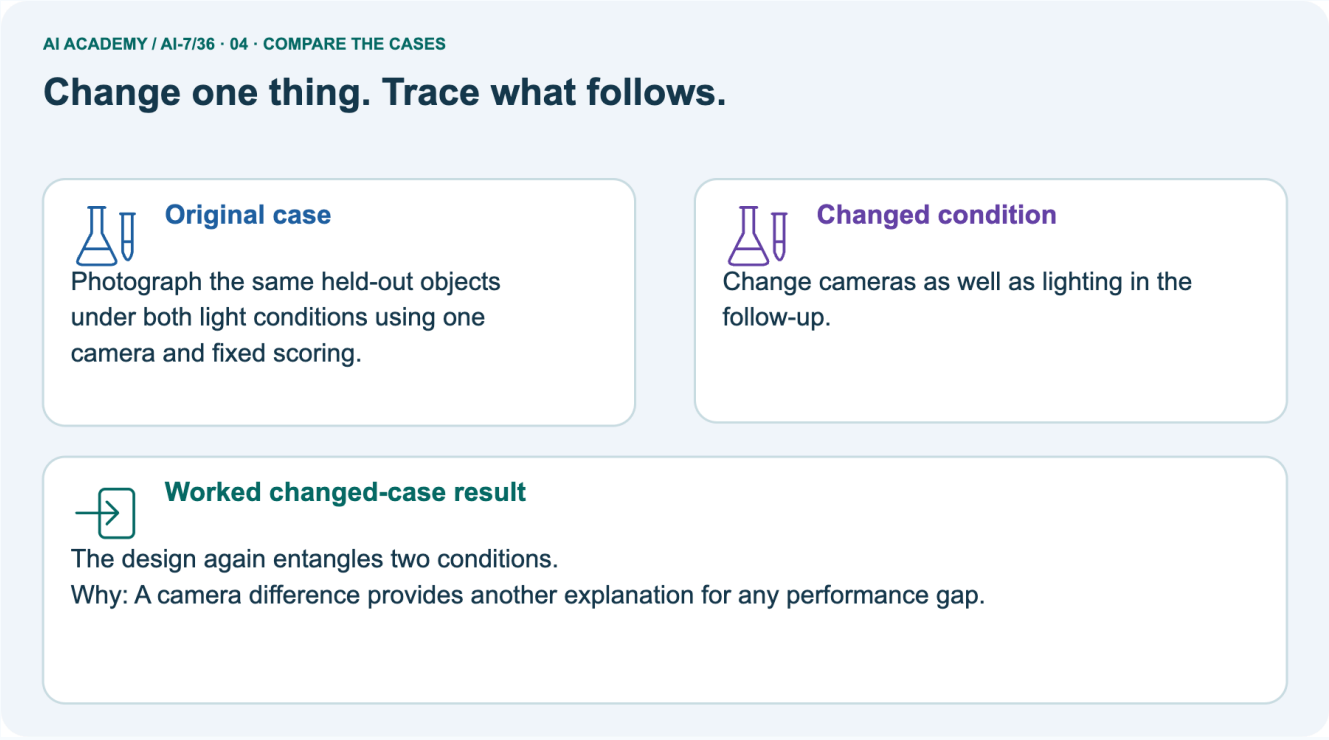
Defence: different condition slices. Units: correct cases out of 10 per slice. Clear: 8. Shaded: 4.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Photograph the same held-out objects under both light conditions using one camera and fixed scoring. Changed condition: Change cameras as well as lighting in the follow-up.

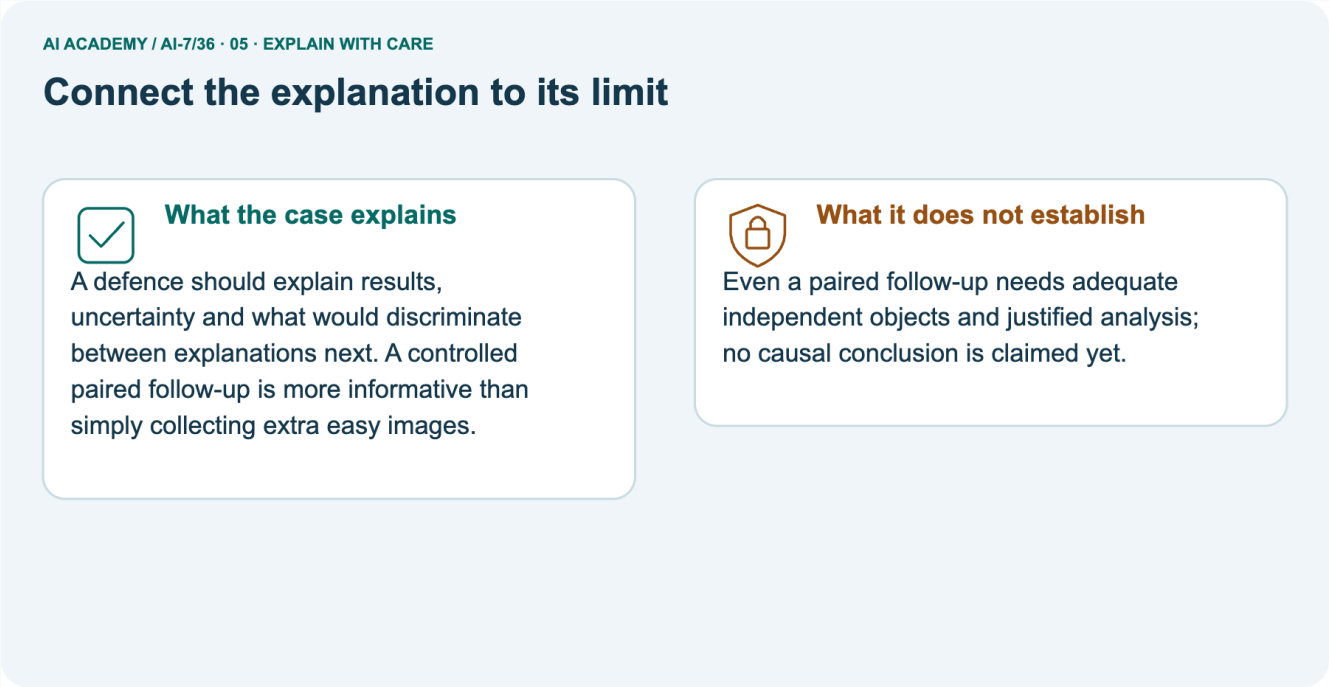
Expected result: The design again entangles two conditions. Reason: A camera difference provides another explanation for any performance gap.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A camera difference provides another explanation for any performance gap.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Limit: Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute the clear and shaded rates and identify alternative explanations.

How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Clear is **80%**; shaded is 40%. Shade sensitivity is plausible, but camera or object differences may also matter.

Photograph the same held-out objects under both light conditions using one camera and fixed scoring. Changed case: The design again entangles two conditions.

A camera difference provides another explanation for any performance gap.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Research defense and next questions: the case

02. Follow the deciding steps

03. Defence: different condition slices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.